

JINMAN ZHAO

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EDUCATION

University of Toronto

Ph.D. in Computer Science

Sept 2019 – Oct 2026

- Research Interest: Post Training, Efficient NLP, Infrastructure, Formal Language Theory

Honors Bachelor of Science in Computer Science

Sept 2014 – Aug 2019

- CGPA: 4.0/4.0

PROFESSIONAL EXPERIENCES

Bytedance Inc.

Senior Researcher (ByteDance Talent Program)

July 2026 - present

Hangzhou, China

- LLM research

Alibaba Inc.

Senior Researcher / Engineer in AI Field Intern (Alibaba Talent Program)

Aug 2025 - Oct 2025

Hangzhou, China

- Post-train LLMs via GRPO/PPO for code/search agent
- Applied LLMs to address challenges in B2B e-commerce (intent recognition, product search, etc)

Baidu Inc.

Research Intern

Mar 2025 - Aug 2025

Beijing, China

- Contributed to the pretraining of a 100B MoE model using Megatron(on policy distillation, post-train)
- Conducted DPO/GRPO research on enhancing mathematical reasoning in LLMs

Huawei Technology Canada

Research Engineer

May 2019 - Dec 2024

Markham, ON

- Built a polyhedral-based framework for loop transformations, improving performance by up to 30% on key machine learning kernels (e.g., Jacobi, 2D/3D Convolution, GEMM), enabling optimized execution in ML workloads; the project contributed to securing **\$1.7M USD** in research funding for the team
- Developed an open-sourced converter to transform TVM's intermediate representation into MLIR, enhancing model compatibility and optimization
- Integrated automatic differentiation into a machine learning framework, supporting advanced neural algorithms and models
- LLM for compiler agent

University of Toronto

Undergraduate Research Assistant (Supervisor: Roger Gross)

Sept 2018 - May 2019

Toronto, ON

- Conducted research on Bayesian Neural Networks (BNN), focusing on reducing the computational complexity in high-dimensional models

University of Toronto

Teaching Assistant

Sept 2015 - present

Toronto, ON

AMD

Software Developer Intern

May 2017 - June 2018

Markham, ON

- Collaborated with cross-functional teams to reproduce, debug, and resolve complex GPU board tool issues, improving tool stability by implementing efficient debugging techniques and memory profiling

AWARDS & SCHOLARSHIP

IVADO Postdoctoral Fellowship

Two-year fellowship hosted at McGill University, \$100,000 per year (declined)

Apr 2026

Best Paper Nominee

International Parallel & Distributed Processing Symposium, IPDPS (Core-A, CCF-B)

July 2025

Rose Sheinin Scholarship

Awarded to the top female student graduating from the University of Toronto

May 2019

Governor General's Silver Medal Nominee

Awarded to the student graduating with the highest average from a University of Toronto

June 2019

First Prize

National Olympiad in Informatics in Provinces (NOIP), Zhejiang Province, China

2008

Third Prize

National Olympiad in Informatics in Provinces (NOIP), Zhejiang Province, China

2010, 2011

1. X. Chen, M. Jin, J. Niu, Y. Yin, **J. Zhao** and 5 others
Rethinking Functional Anisotropy in Circuit and Sheaf Discovery for LLMs
Forty-Third International Conference on Machine Learning (ICML), July 2026
2. C. Lu, **J. Zhao** ✉, J. Li, Y. Wang, C. He
On the Representation Geometry of LoRA Model Merging
Findings of the Association for Computational Linguistics (ACL), July 2026
3. X. Hu and **J. Zhao**
Fin-Bias: Comprehensive Evaluation for LLM Decision-Making under human bias in Finance Domain
Findings of the Association for Computational Linguistics (ACL), July 2026
4. J. Shi, X. Yin, J. Huang and **J. Zhao** ✉
CodeHacker: Automated Test Case Generation for Detecting Vulnerabilities in Competitive Programming
Annual Meeting of the Association for Computational Linguistics (ACL) (*Oral*), July 2026
5. K. Wang, T. Zhou, J. Lei, J. Wang, **J. Zhao** and 8 others
Video-to-Audio Generation with Diffusion Transformer and Semantic-Temporal Alignment-Ranked DPO
The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), June 2026
6. S. Guan, M. Lin, C. Xu, **J. Zhao**, D. Greene
Teaching VLMS to Admit Uncertainty in OCR on Visually Degraded Documents
International Conference on Learning Representations (ICLR), Apr 2026
7. **J. Zhao**, E. Xue, H. Wu, Z. Li, Z. Sun, S. Wang, X. Chen, G. Penn
Beyond Step Pruning: Information Theory Based Step-level Optimization for Self-Refining LLMs
AAAI Conference on Artificial Intelligence (AAAI), Jan 2026
8. **J. Zhao**, X. Zhang, J. Li, J. Niu, Y. Hu, E. Min, G. Penn
Tiny Budgets, Big Gains: Parameter Placement Strategy in Parameter Super-Efficient Fine-Tuning
Empirical Methods in Natural Language Processing (EMNLP), November 2025
9. L. Cao* and **J. Zhao***
Pretraining on the Test Set Is No Longer All You Need: A Debate-Driven Approach to QA Benchmarks
Conference on Language Modeling (COLM), October 2025
10. X. Zhang*, **J. Zhao*** ✉, Y. Zhong, S. Guan, L. Cao, Y. Wang
UORA: Uniform Orthogonal Reinitialization Adaptation in PEFT of Large Models
Annual Meeting of the Association for Computational Linguistics (ACL), July 2025
11. Y. Zhong, **J. Zhao**, Y. Zhou
Low-Rank Interconnected Adaptation across Layers
Annual Meeting of the Association for Computational Linguistics (ACL), July 2025
12. Z. Hu, Y. Liu, **J. Zhao**, S. Wang, Y. Wang, W. Shen, Q. Gu, A. Luu, S. Ng, Z. Jiang and B. Hooi
LongRecipe: Recipe for Efficient Long Context Generalization in Large Language Models
Annual Meeting of the Association for Computational Linguistics (ACL), July 2025
13. S. Guan, M. Lin, C. Xu, X. Liu, **J. Zhao**, J. Fan, Q. Xu, D. Greene
PreP-OCR: A Complete Pipeline for Document Image Restoration and Enhanced OCR Accuracy
Annual Meeting of the Association for Computational Linguistics (ACL), July 2025
14. **J. Zhao**, G. Penn
An Efficient Parser for Bounded-Order Product-Free Lambek Categorical Grammar via Term Graph
The 18th International Conference on Parsing Technologies, July 2025
15. **J. Zhao**, A. Wang, X. Yue and T. S. Abdelrahman
PolyMorphous: A Polyhedral-based Approach to Composable Loop Transformations
International Parallel & Distributed Processing Symposium (IPDPS) (*best paper nominee*), June 2025
16. G. Cai, R. Zhang, H. He, Z. Zhang, D. Ergu, Y. Cao, **J. Zhao**, B. Hu, Z. Liao, Y. Zhao, Y. Cai
Msdet: Receptive field enhanced multiscale detection for tiny pulmonary nodule
IEEE International Conference on Multimedia & Expo (ICME), June 2025
17. **J. Zhao** and G. Penn
Inside-Outside Algorithm for Probabilistic Product-Free Lambek Categorical Grammar
Conference on Computational Linguistics (COLING), January 2025
18. **J. Zhao** and G. Penn
Categorical Grammar Supertagging via Large Language Models
Empirical Methods in Natural Language Processing (EMNLP), November 2024
19. **J. Zhao** and X. Zhang
Large language model is not a (multilingual) compositional relation reasoner
Conference on Language Modeling (COLM), October 2024
20. **J. Zhao** and G. Penn
A generative model for Lambek categorical sequents
Conference on Computational Linguistics (COLING), May 2024

21. **J. Zhao** and G. Penn
A generative process for Lambek categorial proof nets
Meeting on the Mathematics of Language (MOL), December 2021
22. **J. Zhao**, Y. Ding, Z. Qian, Y. Wang
Gender Bias in Large Language Models across Multiple Languages
Proceedings of the 5th Workshop on Trustworthy Natural Language Processing @ NAACL, Apr 2025
23. Y. Wang, **J. Zhao**, and Y. Lawryshyn
GPT-signal: Generative AI for semi-automated feature engineering in the alpha research process
Workshop on the Financial Technology and Natural Language Processing @ IJCAI, August 2024
24. **J. Zhao**, G. Penn and H. Ling
Structural realization with GGNNs
Workshop on Graph-Based Methods for Natural Language Processing @ NAACL, June 2021

SELECTED PREPRINT

25. Y. Quan, Z. Shi, **J. Zhao** and R. Tang
From Plausible to Grounded: Reinforcing Structured Consistency in Video MLLMs
Submitted to COLM 2026
26. X. Zhang, J. Huang, Y. Ding and **J. Zhao** ✉
Fusion R1: Empower PyTorch Graph Fusion via LLM with Reinforcement Learning
Submitted to Neurips 2026
27. X. Mao*, **J. Zhao***, G. Penn, C. Wang
What Happens Inside Agent Memory? Circuit Analysis from Emergence to Diagnosis
Submitted to Neurips 2026
28. C. Yang*, Z. Li*, **J. Zhao***, Z. Lin
LaRA-WM: Test-Time Latent Action Refinement for Robotic Manipulation in Dynamic Environments
Submitted to Neurips 2026
29. Z. Li, **J. Zhao** and 6 others
Responsive Reasoning Evolution via Capability-Adaptive Hint Scaffolding GRPO
Submitted to Neurips 2026
30. H. Wu, E. Min, **J. Zhao** and 7 others
Stability-Aware and Gradient-Efficient Alignment for Reasoning Models
Submitted to EMNLP 2026
31. **J. Zhao** and 5 others
Syntactic Prediction through Reinforcement Learning
Submitted to EMNLP 2026
32. **J. Zhao** and 10 others
IntentEval: Evaluating Whether Large Language Models Answer What Users Actually Mean
Submitted to EMNLP 2026
33. Y. Yang, C. Jin, **J. Zhao** and 6 others
Act More, Decide Less: Skill-Guided Adaptive Action Chunking for Long-Horizon LLM Agents
Submitted to EMNLP 2026
34. Y. Hui, R. Kong, X. Chen, **J. Zhao** and 3 others
VideoNav: Cost-Aware Index Navigation for Long Video Understanding
Submitted to EMNLP 2026
35. Y. Wang*, **J. Zhao***, G. Penn and S. Liu
ATPO: Adaptive Token Aggregation for Policy Optimization
Submitted to AAAI 2026
36. **J. Zhao** and 10 others
CARE-Muon: Conflict-Aware Energy Redistribution via Long- and Short-Term Momentum
Submitted to AAAI 2026
37. J. Huang and 5 others and **J. Zhao** ✉
BoostMerge: Headroom-Weighted Residual Refinement for Model Merging
Submitted to AAAI 2026

PATENT

1. A Polyhedral Schedule Completion Mechanism Respecting User Requirements with a Look-Ahead Optimization (US20250231809).
2. Methods, systems, apparatuses, and computer-readable media for source code loop optimization.

TALK

TVM conference

A TVM IR to MLIR Automatic Converter: Bridging TVM with MLIR Ecosystem

Nov 2020